📝 Unit 1 Guided Notes — The Real Number System

ANSWER KEY

Classify it, root it, estimate it, convert it, and order it. Every number has a home in the real number system.

🎯 By the end of this unit, I can…

- Sort numbers into the real number system (rational vs. irrational)
- Find perfect squares and their roots
- Estimate square roots on a number line and to the nearest tenth
- Convert between fractions and decimals, including repeating decimals
- Compare and order real numbers from least to greatest

Standards: NC.8.NS.1 · NC.8.NS.2

Answer key: every blank and final answer is filled in green.

1 Number Classification

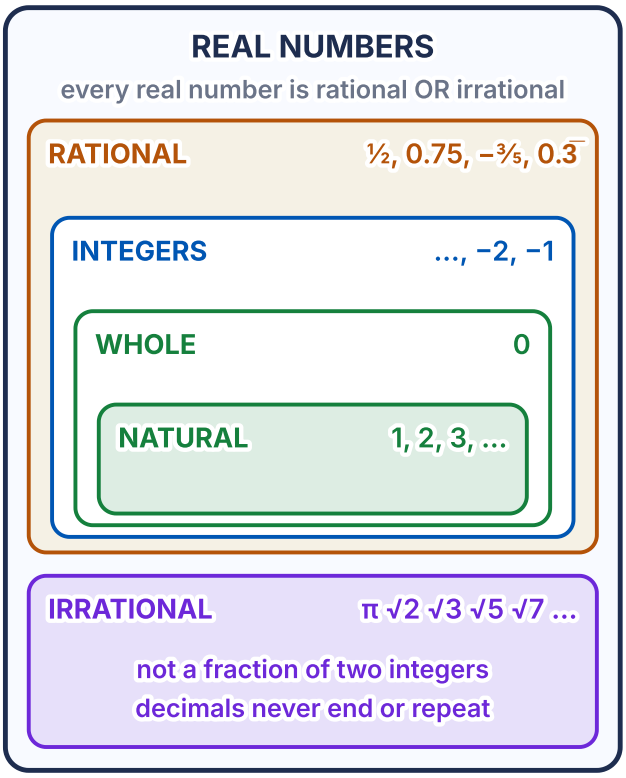
Every real number belongs to one or more of these sets. Fill in the definition for each:

Set	Examples	Definition
Natural Numbers	1, 2, 3, 4, …	***Counting*** numbers
Whole Numbers	0, 1, 2, 3, …	Naturals + ***zero***
Integers	…, −2, −1, 0, 1, 2, …	Whole numbers + their ***negatives***
Rational Numbers	$\frac{1}{2}$, 0.75, $-\frac{3}{5}$, $0.\overline{3}$	Can be written as a ***fraction*** of two integers (includes terminating & repeating decimals)

Irrational Numbers	π, √2, √3	CANNOT be written as a fraction of two integers; ***non-repeating*** , ***non-terminating*** decimal

The Real Number System — Nested Sets

Each set fits *inside* the next one. 5 is natural, whole, an integer, and rational. π is not in any of these boxes.



Natural → Whole → Integers → Rational (each fits inside the next). Irrational numbers sit **outside** the rational box.

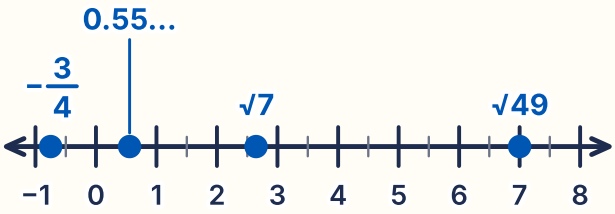
Match the Set to its Definition

MATCH Score 3 **Click a set on the left, then its definition on the right.**

B	1. Natural Numbers	**A.** Whole numbers + their negatives
D	2. Whole Numbers	**B.** Counting numbers: 1, 2, 3, ...
A	3. Integers	**C.** Non-repeating, non-terminating decimal
C	4. Irrational Numbers	**D.** Naturals + zero

Quick Check

MC Score 3 **Which number is *irrational*?**

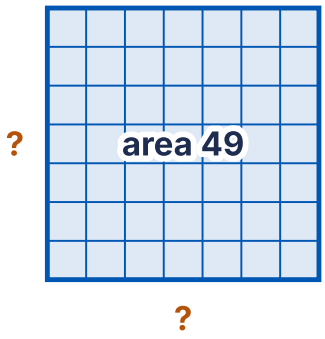


All four choices on a number line. Which one has a decimal that never ends **and** never repeats?

- ○ $0.5\overline{5}$
- ○ √49
- ● **√7**
- ○ $-\frac{3}{4}$

ERROR ANALYSIS **A student claims:**

`"√49 is irrational because it has a square-root symbol."` **What's the mistake?**



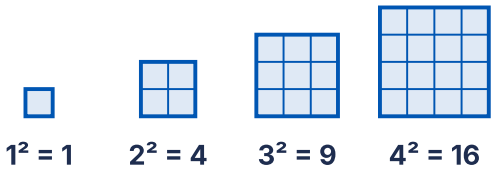
√49 asks: what side length makes a square with area 49? Count the tiles on one side.

The square-root symbol does **not** make a number irrational. √49 = 7, a whole number — so it is **rational**. A root is irrational only when the number underneath is *not* a perfect square (like √7 or √2).

2 Perfect Squares & Square Roots

📌 **THE RELATIONSHIP:** Squaring and square-rooting are INVERSE OPERATIONS — they undo each other. If $5^2 = 25$, then $\sqrt{25} = 5$. Know one, know the other!

A **perfect square** is a whole number times itself ($n \times n = n^2$). Its **square root** takes you back to that whole number. So $\sqrt{}$(a perfect square) is always a whole number.



Perfect squares are numbers of tiles that make a **perfect square**. The side is the square root: $\sqrt{16} = 4$.

Perfect Squares 1–15 — Reference Chart

Keep this chart handy — you'll use it constantly for roots and estimating.

n	n^2	n	n^2	n	n^2
1	1	6	36	11	121
2	4	7	49	12	144
3	9	8	64	13	169
4	16	9	81	14	196
5	25	10	100	15	225

💡 **READ IT BOTH WAYS:** the chart says $12^2 = 144$, which means $\sqrt{144} = 12$. Every row is two facts in one.

Squaring by Hand — the Standard Algorithm

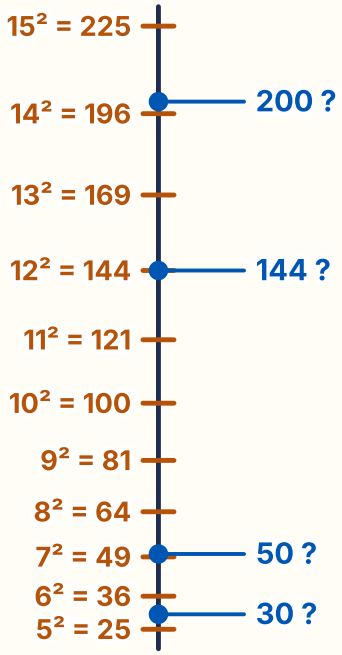
To square a number, multiply it by itself. Line up the digits and multiply.

	1 4	
×	1 4	
	5 6	4 × 14 = 56
+	1 4 0	10 × 14 = 140 (the 0 holds the place)
	1 9 6	Add: 56 + 140 = 196

$14 \times 14 = 196$, so $14^2 = 196$ and $\sqrt{196} = 14$.

Quick Check

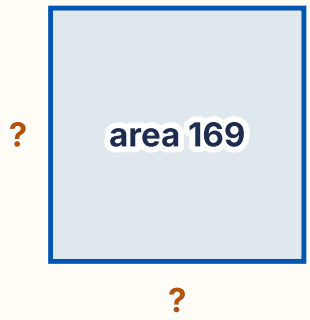
MC Score 3 **Which of these is a *perfect square*?**



Perfect squares from 5^2 to 15^2. A perfect square lands **exactly on** one of the gold marks.

- ○ 30
- ○ 50
- ● **144**
- ○ 200

CALC Score 3 **What is √169 ?**



What side length makes a square with area 169?

👉 **WHAT IF IT'S NOT A PERFECT SQUARE?** Then the root is irrational (like √7 or √20) — it never ends or repeats. We estimate those in Section 3.

3 Estimating Square Roots

📌 **THE BIG IDEA:** When a number is NOT a perfect square, its square root is irrational — it never ends or repeats. We can't write it exactly, but we can *estimate* it. Two tools: a number line, and a step-by-step algorithm.

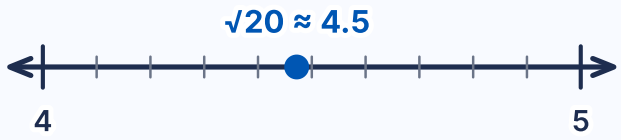
Method 1 — Place it on a Number Line

Great for "between which two whole numbers?" questions.

1. Find the two perfect squares your number lands **between**.
2. Take the √ of each — those are two back-to-back whole numbers. Your root lands between them.
3. Decide which one it's **closer** to: is your number nearer the lower or the upper perfect square?
4. Place it on the number line.

Example: place √20

16 < **20** < **25**, so √20 is between **4** and **5**. Since 20 is closer to 16, √20 sits a little below the middle — closer to 4.



16 < 20 < 25, so √20 is between 4 and 5. (Calculator: 4.47…)

Method 2 — The Estimating Algorithm (to the nearest tenth)

How far is your number into the gap?

Between the two perfect squares, find how far along your number sits. That fraction, added to the **smaller root**, is your estimate. Watch the colors: **gold = perfect squares**, **green = the roots**, **blue = your number**.

Step 1 — Trap your number between two perfect squares.

$$36 < 45 < 49$$

√36 = **6** · √49 = **7** → so √45 is between **6** and **7**. The smaller root is **6**.

Step 2 — Build the fraction: how far in is your number?

$$\frac{\text{your number} - \text{lower square}}{\text{upper square} - \text{lower square}} = \frac{45 - 36 = 9}{49 - 36 = 13} = 9 \div 13 \approx 0.7$$

The fraction is always *squares on top and bottom* — never the roots.

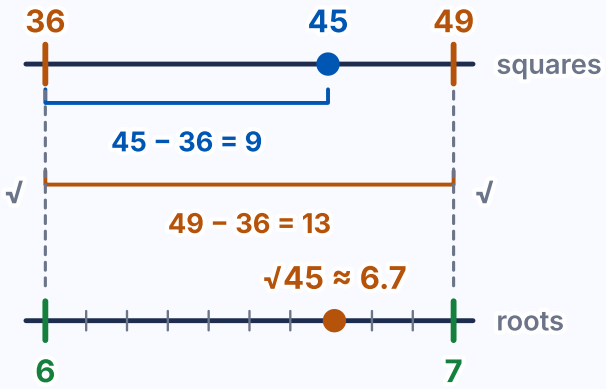
Step 3 — Add the fraction to the smaller root.

$$\sqrt{45} \approx 6 + 0.7 = 6.7$$

Calculator check: √45 = 6.708... ✓

Check by multiplying: 6.7 × 6.7 = 44.89, very close to 45.

```
   6.7
 × 6.7
 -----
   469      7 × 67 = 469
+ 4020      60 × 67 = 4020 (the 0 holds the place)
 -----
  44.89     Add: 469 + 4020 = 4489. 2 decimal places in the factors, so 2 in the answer.
```

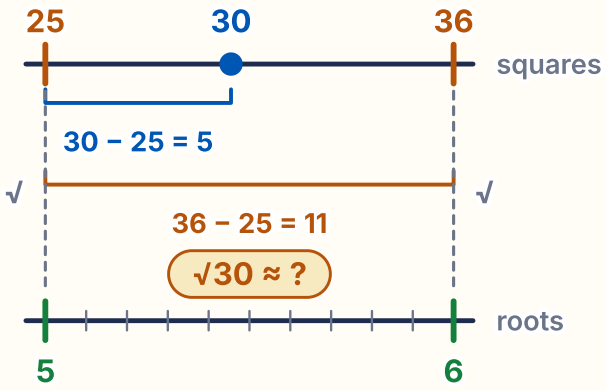


45 is **9 of the 13** steps from 36 to 49, so √45 is about 9 ÷ 13 ≈ 0.7 of the way from 6 to 7.

💡 **IN SHORT:** √n ≈ (smaller root) + (n − lower square) ÷ (upper square − lower square). This is an *estimate* — it can be off by about a tenth from a calculator, and that's fine.

⚠ **DON'T MIX THEM UP:** the number you **add** is a **root** (6). The numbers **inside the fraction** are the **perfect squares** (36 and 49). Roots to add, squares to subtract.

CALC Score 4 **Estimate √30 to the nearest tenth using the algorithm.** (Hint: 25 < 30 < 36)



Use the two gaps to build your fraction, then add it to 5.

25 < 30 < 36 → roots 5 and 6. Fraction = (30 − 25) ÷ (36 − 25) = 5 ÷ 11 ≈ 0.45. So √30 ≈ 5 + 0.45 = 5.45 → **5.5**. (Calculator: 5.477...)

Negative Square Roots

📌 **JUST FLIP THE SIGN:** a negative in front puts your estimate on the other side of zero. Estimate the positive root first, then make it negative. Example: −√40 → √40 ≈ 6.3, so −√40 ≈ −6.3.



Same distance from 0, opposite sides: √40 ≈ 6.3 and −√40 ≈ −6.3.

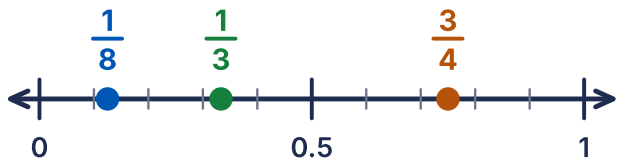
🧭 **WHICH METHOD?** Use the **number line** for "between which whole numbers." Use the **algorithm** when you need a decimal to the nearest tenth. Both start the same way: find the two perfect squares that surround your number.

4 Converting Fractions ↔ Decimals

Fraction → Decimal

Divide the numerator by the denominator (top ÷ bottom).

Fraction	Division	Decimal
$\frac{3}{4}$	3 ÷ 4	0.75
$\frac{1}{8}$	1 ÷ 8	0.125
$\frac{1}{3}$	1 ÷ 3	0.333… = $0.\overline{3}$



Each fraction is a point between 0 and 1. Dividing top ÷ bottom tells you its decimal name.

Worked Example — Divide 5 ÷ 6 with Long Division

Put 5 inside the bracket and 6 outside. Write a decimal point and zeros after the 5. The gold digit is the 0 you bring down.

```
    0.8 3 3
6 | 5.0 0 0
  − 4 8          6 × 8 = 48
    2 0          50 − 48 = 2. Bring down a 0.
  − 1 8          6 × 3 = 18
      2 0        20 − 18 = 2. Bring down a 0.
    − 1 8        6 × 3 = 18
        2        20 − 18 = 2. Same remainder again, so the 3 repeats.
```

$\frac{5}{6}$ = 0.8333... = $0.8\overline{3}$. Only the 3 repeats, so the bar goes over the 3.

💡 **KEY INSIGHT:** Repeating decimals are ***RATIONAL*** because they can be written as fractions. Non-repeating, non-terminating decimals (like π) are ***IRRATIONAL***. Use a **bar** over the digits that repeat: $0.\overline{3}$ means 0.3333...

```
    0.6 6 6
3 | 2.0 0 0
  − 1 8          3 × 6 = 18
    2 0          20 − 18 = 2. Bring down a 0.
  − 1 8          3 × 6 = 18
    2 0          20 − 18 = 2. Bring down a 0.
  − 1 8          3 × 6 = 18
      2          20 − 18 = 2. The remainder is 2 again, so the 6 repeats.
```

Why 2 ÷ 3 repeats: every step leaves remainder **2**, so the same digit (6) comes out forever.

CALC Score 4 **Use long division to write $\frac{7}{11}$ as a decimal. Watch for a two-digit pattern.**

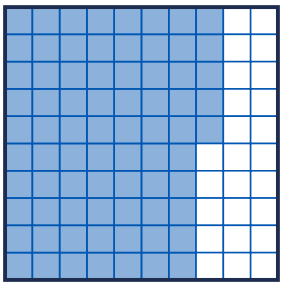
```
     0.6 3 6 3
11 | 7.0 0 0 0
   − 6 6           11 × 6 = 66
     4 0           70 − 66 = 4. Bring down a 0.
   − 3 3           11 × 3 = 33
     7 0           40 − 33 = 7. Bring down a 0.
   − 6 6           11 × 6 = 66
     4 0           70 − 66 = 4. Bring down a 0.
   − 3 3           11 × 3 = 33
       7           40 − 33 = 7. Back to 7, where we started, so 6, 3 repeats.
```

$\frac{7}{11}$ = 0.6363... = $0.\overline{63}$. The bar goes over both digits, 6 and 3.

Decimal → Fraction (terminating)

Read it, write it over its place value, then simplify — proof that every terminating decimal is **rational**.

Decimal	Read / write	Fraction (simplest form)
0.75	75 hundredths → $\frac{75}{100}$	= *3/4*
0.6	6 tenths → $\frac{6}{10}$	= *3/5*
0.125	125 thousandths → $\frac{125}{1000}$	= *1/8*



0.75 = 75 of 100



0.6 = 6 of 10

1s	10ths	100ths	1000ths
0	1	2	5

0.125 = 125 thousandths

Read the place value: that is the fraction before you simplify.

Repeating Decimal → Fraction: The 9's Rule

📌 **THE 9's RULE:** The repeating digits go in the NUMERATOR. The DENOMINATOR is made of 9's — one 9 for each repeating digit.

Repeating digits	Denominator	Example	Simplified
1 digit	*9*	$0.\overline{3} = \frac{3}{9}$	= *1/3*
2 digits	*99*	$0.\overline{27} = \frac{27}{99}$	= *3/11*
3 digits	*999*	$0.\overline{125} = \frac{125}{999}$	*125/999*



Bracket one repeating block and count its digits. That count tells you how many 9s.

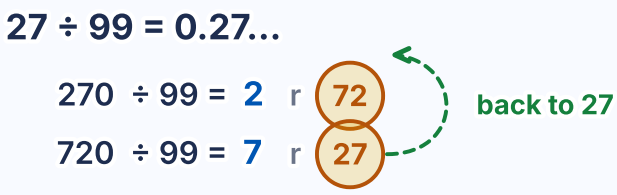
Worked Example — Convert $0.\overline{27}$ (0.272727…) to a fraction

Step 1 — Count the repeating digits: 2 digits repeat (the block "27").

Step 2 — Denominator = that many 9's: 2 digits → 99.

Step 3 — Numerator = the repeating block: $\frac{27}{99}$.

✓ **Step 4 — Simplify:** $\frac{27}{99} = \frac{3}{11}$.



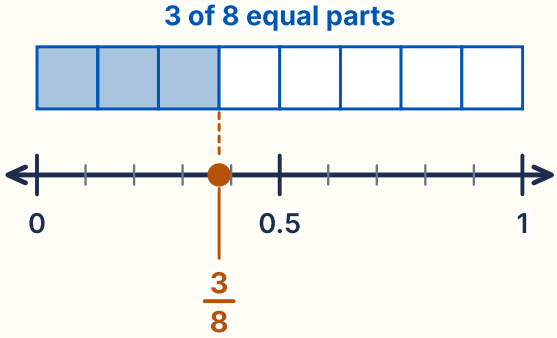
Check: 27 ÷ 99 gives 0.2727... back

Check the fraction: 27 ÷ 99 brings back the same remainder, 27, so the digits 2, 7 repeat.

⚠ **IMPORTANT:** Use the 9's rule when the repeat starts *right after* the decimal point, like 0.272727... Always simplify your fraction: $\frac{27}{99} = \frac{3}{11}$.

Quick Check

MC Score 3 **Which decimal equals the fraction $\frac{3}{8}$?**



$\frac{3}{8}$ is 3 of 8 equal parts. Divide 3 ÷ 8 to name the point as a decimal.

- ○ 0.38
- ● **0.375**
- ○ 0.83
- ○ 0.625

```
    0.3 7 5
  ┌─────────
8 │ 3.0 0 0
  - 2 4          8 × 3 = 24
  ─────
    6 0          30 − 24 = 6. Bring down a 0.
  - 5 6          8 × 7 = 56
  ─────
      4 0        60 − 56 = 4. Bring down a 0.
    - 4 0        8 × 5 = 40
    ─────
        0        40 − 40 = 0. Nothing is left, so the decimal ends.
```

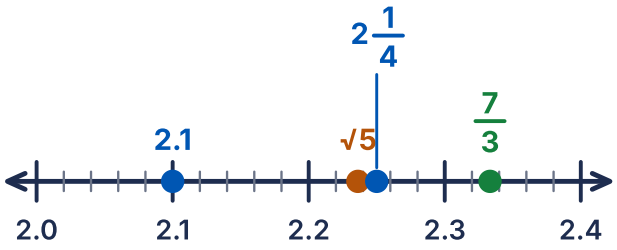
5 Ordering Numbers Least to Greatest

📌 **THE STRATEGY:** To compare a mix of fractions, decimals, mixed numbers, and roots — turn *everything* into a decimal first. Then ordering is easy.

Worked Example — Order $\sqrt{5}$, $2\frac{1}{4}$, $\frac{7}{3}$, 2.1

Number	Decimal	How?
$\sqrt{5}$	≈ *2.24*	Estimating algorithm (between 4 and 9)
$2\frac{1}{4}$	= *2.25*	2 + (1 ÷ 4) = 2.25
$\frac{7}{3}$	≈ *2.33*	7 ÷ 3 = 2.333...
2.1	= *2.1*	Already a decimal

Order from least to greatest: *2.1* < *√5* < *2¼* < *7/3*



Least → Greatest: **2.1 < √5 < $2\frac{1}{4}$ < $\frac{7}{3}$**

💡 **MIXED NUMBERS:** turn them into a decimal — whole number + (numerator ÷ denominator). Example: $3\frac{2}{5}$ = 3 + (2 ÷ 5) = 3 + 0.4 = 3.4.

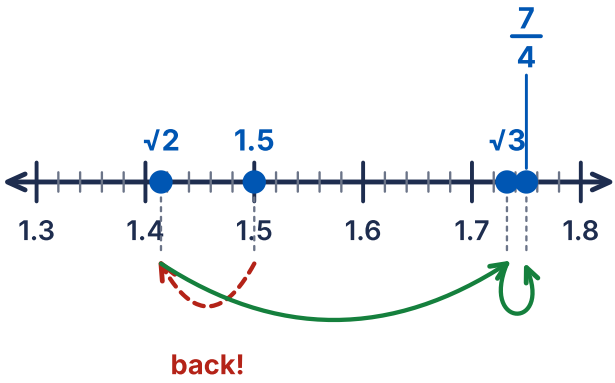
⚠ **WATCH OUT:**

- Give every number the same number of decimal places (3.5 → 3.500). Then 3.375 < 3.500 is easy to see.
- Estimate each root first (√5 ≈ 2.24).

ERROR ANALYSIS To order 1.5, √2, $\frac{7}{4}$, √3, a student wrote $1.5 < \sqrt{2} < \sqrt{3} < \frac{7}{4}$.

Are they right?

Convert first: 1.5 = 1.5, √2 ≈ 1.41, $\frac{7}{4}$ = 1.75, √3 ≈ 1.73. The student forgot that √2 ≈ 1.41 is the *smallest*. Correct order: **$\sqrt{2} < 1.5 < \sqrt{3} < \frac{7}{4}$**.



Follow the student's order. Least to greatest should always hop **right**; their first hop goes **back**.

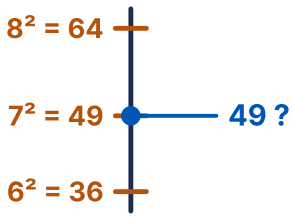
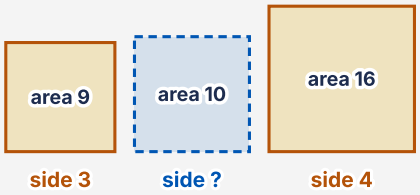
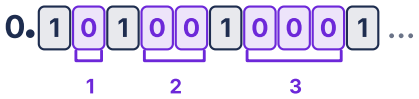
✏ PART 2 — Practice Problems

Show all work on your printed copy. Circle your final answer, then click to check on screen.

A Classify Each Number

Rational = can be written as a fraction (includes terminating & repeating decimals).
Irrational = non-repeating, non-terminating.

#	Classification
Score 3 1. 0.333...  The 3 repeats forever.	*Rational (= 1/3)*

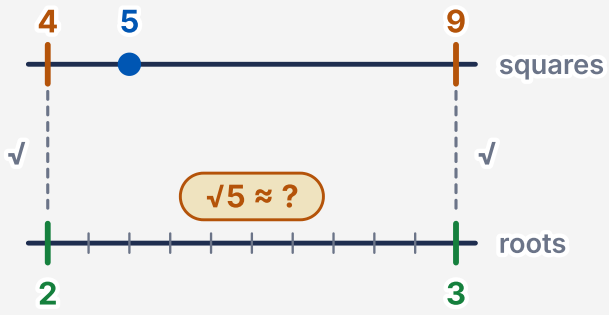
Score 3 2. √49  Does 49 land on a perfect square?	*Rational (√49 = 7)*
Score 3 3. √10  10 is not a perfect square: its side is between 3 and 4.	*Irrational*
Score 4 4. 0.101001...  The runs of 0s grow: 1, 2, 3, ... no block ever repeats.	*Irrational (pattern grows, never repeats)*
Score 4 5. −√81  81 = 9 × 9. −√81 is the opposite of √81.	*Rational (= −9)*

B Estimate Each Square Root (nearest tenth)

Find the perfect squares just below and just above, then estimate. A leading − just flips the sign.

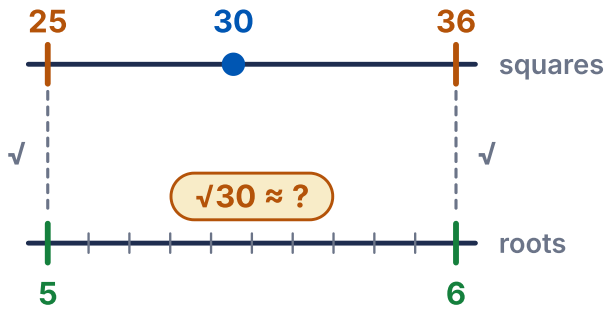
#	Estimate

Score 3 1. √5



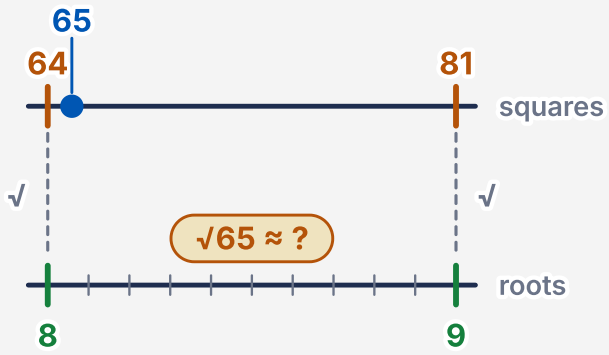
≈ 2.2

Score 4 2. √30



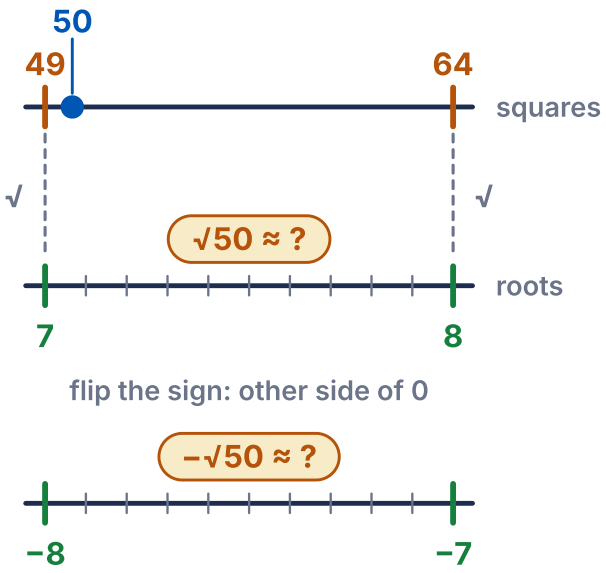
≈ 5.5

Score 4 3. √65



≈ 8.1

Score 5 4. $-\sqrt{50}$



flip the sign: other side of 0

$\approx$ -7.1

C Order from Least to Greatest

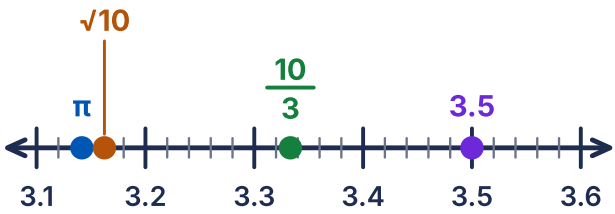
Convert each number to a decimal, then arrange from least to greatest. Show your decimal conversions!

Score 3 **1. Order: π, √10, $\frac{10}{3}$, 3.5**



Turn each number into a decimal, then place it on this line: π, √10, $\frac{10}{3}$, 3.5.

Decimals: 3.14, 3.16, 3.33, 3.5 → **π < √10 < $\frac{10}{3}$ < 3.5**

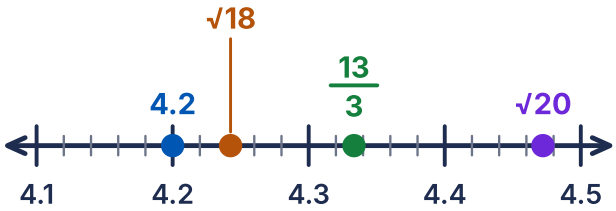


Score 4 **2. Order: 4.2, √18, $\frac{13}{3}$, √20**

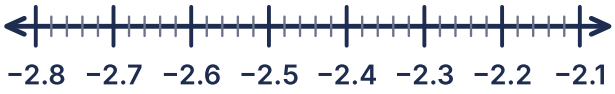


Turn each number into a decimal, then place it on this line: 4.2, √18, $\frac{13}{3}$, √20.

Decimals: 4.2, 4.24, 4.33, 4.47 → **4.2 < √18 < $\frac{13}{3}$ < √20**

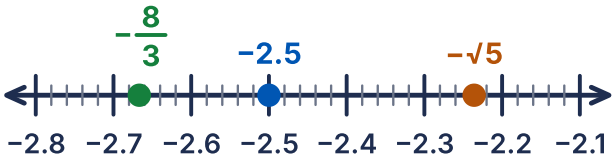


Score 5 **3. Order: −2.5, −√5, $-\frac{8}{3}$**



Turn each number into a decimal, then place it on this line: −2.5, −√5, $-\frac{8}{3}$.

Decimals: −2.5, −2.24, −2.67 → **$-\frac{8}{3}$ < −2.5 < −√5**



D Explain Your Thinking

These are Score 5 questions. Pick the answer you can **defend**. Change numbers to decimals when you need to.

Score 5 **Which statement about the real number system is TRUE?**

- ○ Every rational number is an integer.
- ● **Every integer is a rational number.**
- ○ Some numbers are both rational and irrational.
- ○ Every square root is irrational.

Score 5 **Why is $\frac{5\pi}{2}$ irrational, even though it is written as a fraction?**

- ○ It is not irrational. Every fraction is rational.
- ○ Because 2 does not divide 5 evenly.
- ● **The top, 5π, is not an integer.**
- ○ Because its decimal has more than 10 digits.

Score 5 **Which list is in order from least to greatest?**

- ○ π, $\sqrt[3]{30}$, $\sqrt{10}$, 3.2
- ○ 3.2, $\sqrt{10}$, π, $\sqrt[3]{30}$
- ● **$\sqrt[3]{30}$, π, $\sqrt{10}$, 3.2**
- ○ $\sqrt{10}$, $\sqrt[3]{30}$, π, 3.2